

From readings to predictions

A multi-model pipeline screens four patterns and flags unusual sensor windows.

01 SENSOR INPUT

Heart rate • SpO2 • Temperature • Ambient light • GPS

CSV: pet_id + timestamp + six sensor columns; target labels are not required.

02 PREPARE EACH PET'S HISTORY

Validate → clean → GPS speed + baseline differences → overlapping windows

15 readings per window; advance 5 readings. The UI expects one reading per minute.

46 window summary features

Means, slopes, extremes, activity fractions

15 × 10 sequence values

Ordered sensor and derived measurements

XGBOOST

4 classifiers
300 trees each
One per condition

ISOLATION FOREST

Healthy windows only
23 selected features
Anomaly percentile

LSTM

Shared 2-layer encoder
Learns time sequences
4 output scores

TCN

Temporal convolutions
Dilated residual blocks
4 output scores

03 FOUR TASK-SPECIFIC ENSEMBLES

Each logistic-regression model combines XGBoost + LSTM + TCN + anomaly.

The anomaly signal also remains a separate output for alerts and risk scoring.

04 RESULTS FOR EACH WINDOW

4 condition estimates + anomaly → alerts → smoothed health score (0-100)

CODE MAP

streamlit_app.py → ui_data.py / features.py → inference.py + models.py
health_score.py creates the score; pipeline.py trains and saves artifacts.

How to read the results

Condition alerts and overall health status use different calculations.

ALERT WHEN SCORE $\geq$ THRESHOLD

Low oxygen	0.10
Fever pattern	0.10
Resting heart rate	0.32
Reduced activity	0.18
General anomaly	0.89

Saved thresholds: artifacts/task_thresholds.json
Selected for validation F1; not clinical cutoffs.

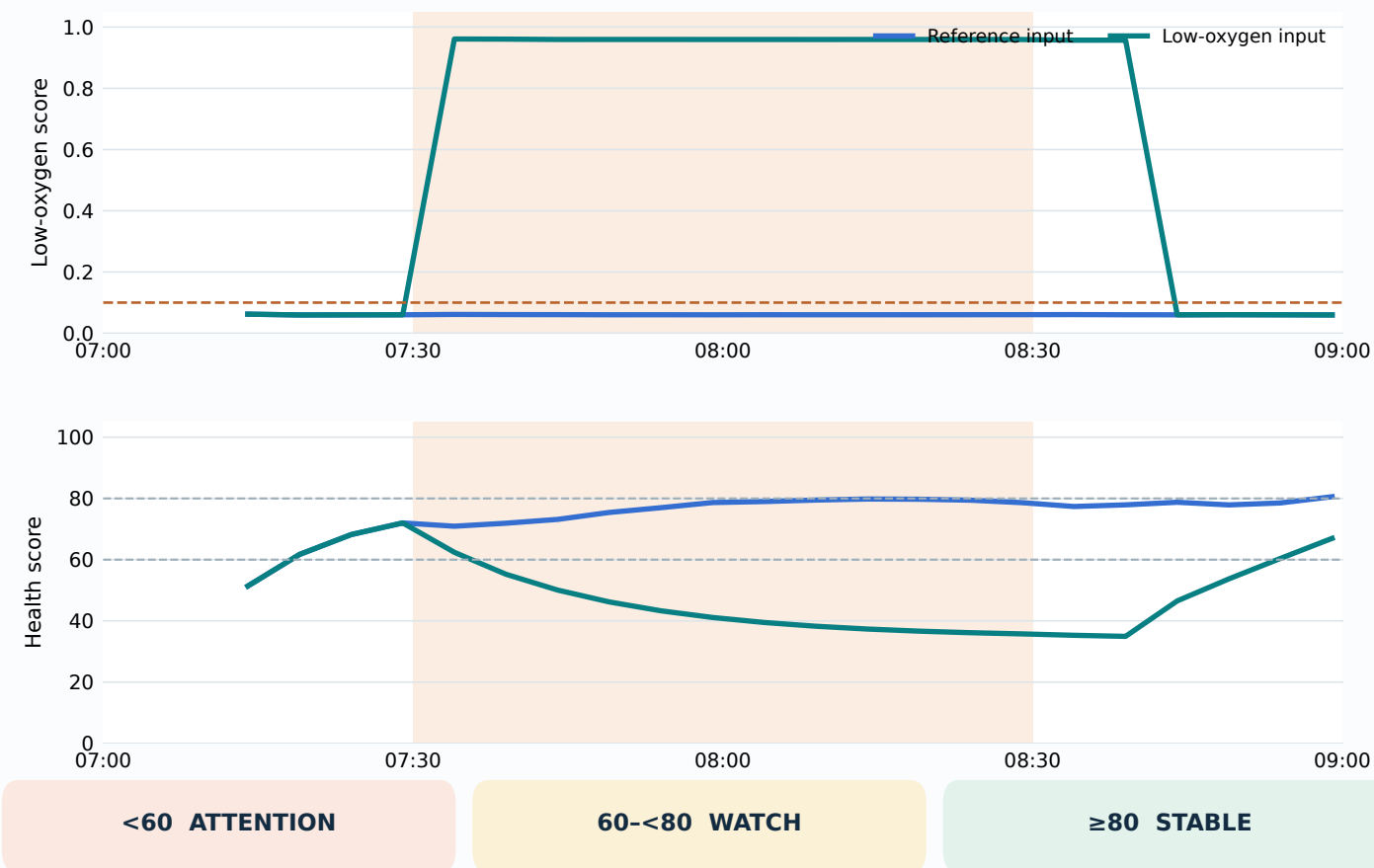
HEALTH SCORE

W = weighted mean of the five signals
Instant risk = $0.70 W + 0.30 \max(\text{signal})$
Smoothed risk = $0.25 \text{ current} + 0.75 \text{ prior}$
Health score = $100 \times (1 - \text{smoothed risk})$

Weights: oxygen .28 / fever .24 / HR .20
activity .12 / anomaly .16. First risk initializes
the smoother; history is separate for each pet.

ACTUAL SAVED-MODEL OUTPUT • SYNTHETIC DEMO

Shaded interval: low-oxygen input episode, 07:30–08:29. Lines end at window timestamps.



Interpretation: anomaly is a percentile, not disease probability. Reference inputs may still trigger alerts. These charts demonstrate software behavior, not real-pet accuracy.

TRAINING / EVALUATION

Saved run: 8 synthetic pets split 5 train / 2 validation / 1 test. Evaluation reports precision, recall, F1 and AUC. Synthetic performance does not establish clinical validity.

Test scenarios & walkthrough

Sample pack: data/test_samples/ • Six simulated pets • No target labels needed



PER PET: 120 READINGS → 22 WINDOWS

07:00 Reference	07:30–08:29 Injected episode	08:30 Recovery
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SCENARIO FILE	WHAT TO LOOK FOR DURING THE EPISODE
01_normal_pattern.csv	No injected condition; inspect false alerts.
02_low_oxygen.csv	SpO2 drops 7 points; inspect oxygen scores.
03_fever_pattern.csv	Temperature +1.15°C; inspect fever scores.
04_high_resting_hr.csv	High HR with low movement; inspect HR scores.
05_reduced_activity.csv	Low morning movement; inspect activity scores.
06_combined_patterns.csv	Oxygen + fever signals can rise together.

ADDITIONAL INPUT CHECKS

07_multiple_pets.csv	720 rows / 6 pets. Switch pets, then rerun.
08_missing_values.csv	Interpolation warning; prediction completes.
09_minimal_columns.csv	Baseline fallback warning; compare scores.

INVALID FILES SHOULD STOP PREDICTION

invalid/too_few_readings.csv → fewer than 15 rows
invalid/missing_spo2_column.csv → missing sensor column
invalid/sampling_gap.csv → timestamps not one minute apart

VERIFIED AGAINST THE CURRENT SAVED MODELS

All 9 valid files produced finite scores and the expected window counts. All 3 invalid files were rejected. This checks execution and validation, not detection accuracy.

Review the full timeline: the final readings are in recovery. Confirm that switching pets changes the result ID. One invalid pet must not hide other pets. Scoring is per batch; this app does not connect to a live device or retain risk across separate uploads.

RUN: `.\.venv\Scripts\python.exe -m streamlit run streamlit_app.py`